Signal to Story: SOC Detection Lab

BLUE TEAM / Endpoint telemetry, alert validation, and incident write-ups

What I set out to do

I set up a small SOC lab to follow a security event from the endpoint to the SIEM. The point is to generate activity in a controlled environment, check whether the logs arrive, tune detections, and write a clear incident report that another analyst can understand.

Scope: a private three-machine lab. Kali generates controlled test activity, Windows produces endpoint telemetry, and Wazuh collects and displays events. These examples are for an isolated lab, not for systems you do not own.

How the project works

Install Wazuh on Ubuntu, onboard the Windows endpoint, enable Sysmon logging, then run one controlled scenario at a time. For every scenario, save the raw event, the alert details, the detection rule, and a short analyst conclusion. Re-run the test after tuning to confirm the rule behaves as expected.

Environment / scope

Kali attacker: 192.168.56.30. Windows test endpoint: 192.168.56.50. Ubuntu/Wazuh SIEM: 192.168.56.40. The Wazuh agent forwards Windows and Sysmon events to the SIEM. Keep the network isolated and use test accounts.

Evidence and findings

These evidence panels are based on the supplied guide's sample SOC scenarios. They show what to capture and what to write down. The actual alert count, timestamps, and rule firing must come from your own Wazuh dashboard and endpoint logs.

01 · Endpoint telemetry health

Capture the Wazuh Agents page showing the Windows test endpoint as Active, plus a recent Sysmon event in Security Events.

```
Evidence to save: wazuh-agent-status.png; sysmon-event-export.json; timestamp and agent name.
```

02 · Port-scan signal

Run an authorized scan against the Windows lab VM, then inspect network-connection telemetry. Confirm the source address and time window match the test.

```
Example scenario: source 192.168.56.30 → lab endpoint 192.168.56.50. Record observed events; do not
assume every scan triggers a built-in rule.
```

03 · Failed-login pattern

Use a controlled test to generate failed logons on a lab account. Windows Security Event ID 4625 is a useful event to review. Verify the custom rule's actual match conditions against the raw event fields.

```
Evidence to save: filtered Event ID 4625 records, Wazuh alert detail, rule ID, target account
(redacted), and the exact test window.
```

04 · Credential-access simulation

If using a safe Atomic Red Team test, capture the test name and ID, endpoint telemetry, corresponding Wazuh event, and whether the alert was a true positive.

```
MITRE ATT&CK example from the guide: T1003.001 — OS Credential Dumping: LSASS Memory. Treat the result as unconfirmed until the alert and raw event are visible.
```

What the evidence means

A useful detection is more than an alert screenshot. The write-up should show why the event looked suspicious, which telemetry supported that conclusion, whether the detection fired as expected, and what an analyst would do next. If the rule did not fire, document that as a tuning gap instead of hiding it.

Recommended next steps

• Confirm the Windows agent is Active and that recent Sysmon events appear in Wazuh before testing.

• Use a dedicated test account and controlled, low-volume scenarios.

• Validate each detection with both a positive test and a benign comparison.

• Record alert time, source and destination, event IDs, rule ID, severity, and MITRE ATT&CK; mapping.

• Document containment steps and lessons learned, but do not claim containment was performed unless you did it.

• Export sanitized screenshots and raw logs; hide usernames, IPs, tokens, and other sensitive details before publishing.

Incident record example

Field	What to record
Incident ID	INC-001 or another unique identifier
Detection time	Actual timestamp from the SIEM, including timezone
Source / target	Lab source IP, endpoint name, and relevant account (redacted for public release)
Evidence	Wazuh alert export, raw Windows/Sysmon events, screenshot, rule ID
Assessment	True positive / false positive / inconclusive, with a short reason
Response	Proposed containment and follow-up; mark clearly if not executed

Evidence integrity note

This portfolio document is based on the supplied project guide. The evidence snippets are illustrative lab examples or expected observations, not proof that a live system was tested by me. To present this as completed work, replace each example with a dated screenshot or exported log from your own lab, keep the original file, and record the tool version, host, timestamp, and test conditions.